

```

a = -113;
b = 113;
n= 1000;
noise = a + (b-a) * rand(n ,1) ;
figure
histogram (noise)

R = 113;
B = 11;
variance = R;
mean = B
std_dev = sqrt (variance);
N = 1000;
noise = std_dev*randn(N,1) + mean;
figure
histogram (noise);

```

```

function [f] = factorial(x)
if (x <= 1)
f=1
else
f=x*factorial(x-1)
end

% using factorial.m
x = input ('factorial number:');
factorial (x)

```

```

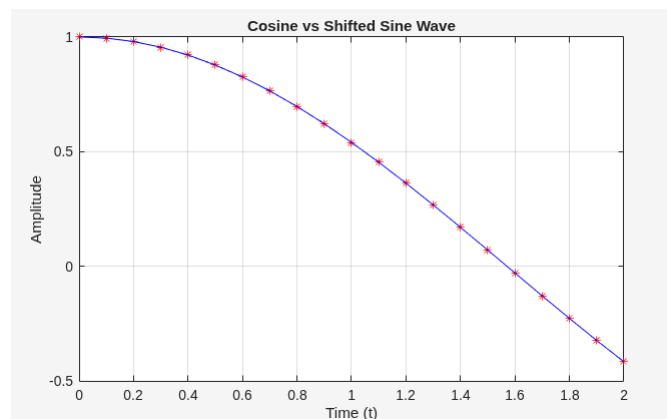
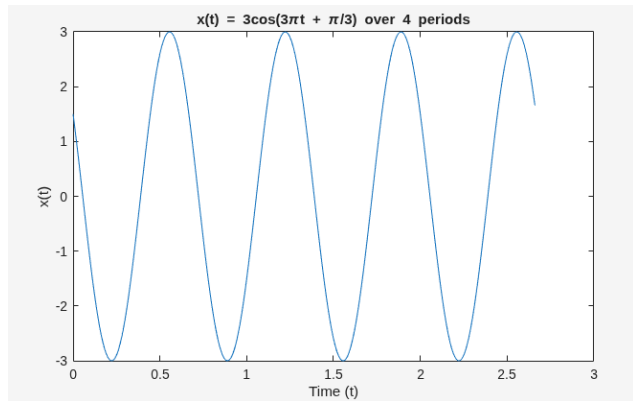
t = 0:0.1:2 ;
x1 = cos(t);
x2 = sin (t+(pi/2));
plot (t , x1 , "r*")
hold
plot (t , x2 , "b-")

```

```

fp = 2/3
% 4 periods
t = 0:0.01:4*fp;
% x(t)
x_t = 3 * cos((3*pi*t)+(pi/3))
plot (t , x_t)
% Labels
xlabel('Time (t)');
ylabel('x(t)');
title('x(t) = 3 cos (3 \pit + \pi/3)
over 4 periods');

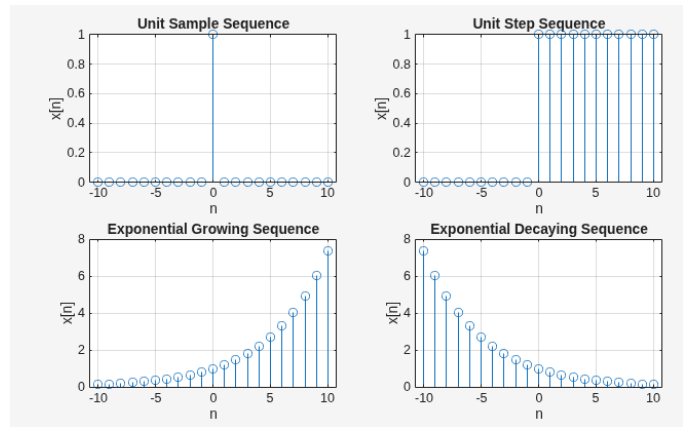
```



```

n1 = -10; n2 = 10; n0 = 0;
n = n1:n2;
% Call functions
x_imp = impseq(n, n0);
x_step = stepseq(n, n0);
x_grow = expgrow(n, 0.2, n0);
x_decay = expdecay(n, 0.2, n0);
% Plot all in one figure
subplot(2,2,1); stem(n, x_imp);
title('Unit Sample Sequence');
xlabel('n'); ylabel('x[n]'); grid on;
subplot(2,2,2); stem(n, x_step);
title('Unit Step Sequence');
xlabel('n'); ylabel('x[n]'); grid on;
subplot(2,2,3); stem(n, x_grow);
title('Exponential Growing Sequence');
xlabel('n'); ylabel('x[n]'); grid on;
subplot(2,2,4); stem(n, x_decay);
title('Exponential Decaying Sequence');
xlabel('n'); ylabel('x[n]'); grid on;

```



```

function x = impseq(n, n0)
% x = gausspuls(n-n0);
x = double(n == n0); %  $\delta[n-n_0]$ 
end

function x = stepseq(n, n0)
%x = double heaviside(n-n0);
% ensure value at n0 is exactly 1
%x(n == n0) = 1;
x = double(n >= n0); %  $u[n-n_0]$ 
end

function x = expgrow(n, a, n0)
% exponential growth
x = exp(a * (n - n0));
end

function x = expdecay(n, a, n0)
% exponential decay
x = exp(-a * (n - n0));
end

```